

Example for scenario making

```
library(tidyverse)
library(AquacropOnR)
```

Meteo data

Here we provide an example for a weather data set:

- TEMP_MAX is the maximum daily temperature in deg C
- TEMP_MIN is the minimum daily temperature in deg C
- RELATIVE_HUMIDITY is the average relative humidity in %
- WIND_SPEED is the average daily wind speed in m/s
- PRECIP_QUANTITY is the total daily precipitation in mm
- GLOBAL_RADIATION is expressed in kWh/m².

```
Meteo_data <- read_delim(paste0(datapath, "data.csv"), delim = ",", show_col_types = F)
head(Meteo_data)
#> # A tibble: 6 x 9
#>   YEAR MONTH DAY TEMP_MAX TEMP_MIN RELATIVE_HUMIDITY WIND_SPEED
#>   <dbl> <dbl> <dbl>   <dbl>   <dbl>         <dbl>       <dbl>
#> 1  2020     1     1     2.9     1.8           95.7         3.1
#> 2  2020     1     2     9.5     1.5           92.8         4
#> 3  2020     1     3    10.4     3           88.9         5.4
#> 4  2020     1     4     8.8     3.3           85.8         3.5
#> 5  2020     1     5     7.5     3.5           87          2.9
#> 6  2020     1     6     8.6     5.2           81.2         3.8
#> # i 2 more variables: PRECIP_QUANTITY <dbl>, GLOBAL_RADIATION <dbl>
```

This data set has some missing data. We solve this here by using the `approxfun` function to interpolate missing values.

```
Meteo_data <- Meteo_data %>%
  mutate(
    DATE = as_date(paste(YEAR, MONTH, DAY, sep = "-"))
  )

START <- "2020-01-01"
STOP <- "2020-12-31"

Tmax_fun <- approxfun(Meteo_data$DATE, Meteo_data$TEMP_MAX)
Tmin_fun <- approxfun(Meteo_data$DATE, Meteo_data$TEMP_MIN)
RH_fun <- approxfun(Meteo_data$DATE, Meteo_data$RELATIVE_HUMIDITY)
Wind_fun <- approxfun(Meteo_data$DATE, Meteo_data$WIND_SPEED)
Precip_fun <- approxfun(Meteo_data$DATE, Meteo_data$PRECIP_QUANTITY)
Radiation_fun <- approxfun(Meteo_data$DATE, Meteo_data$GLOBAL_RADIATION)

Alldays <- seq.Date(from = as_date(START), to = as_date(STOP), by = 1)

Meteo_data <- tibble(
  YEAR = lubridate::year(Alldays),
  MONTH = lubridate::month(Alldays),
  DAY = lubridate::day(Alldays),
  TEMP_MAX = Tmax_fun(Alldays),
  TEMP_MIN = Tmin_fun(Alldays),
  RELATIVE_HUMIDITY = RH_fun(Alldays),
  WIND_SPEED = Wind_fun(Alldays),
  PRECIPITATION = Precip_fun(Alldays),
  GLOBAL_RADIATION = Radiation_fun(Alldays),
  DATE = Alldays
)
```

As AquaCrop requires reference evapotranspiration as an input, we calculate this using the `ETo_calc()` function available through the `MeteoTools` package, based on the `Evapotranspiration` package.

```
library(MeteoTools)

ETo_data <- ETo_calc(
  DATE = Meteo_data$DATE,
  T_max = Meteo_data$TEMP_MAX,
  T_min = Meteo_data$TEMP_MIN,
  RH_max = pmin(Meteo_data$RELATIVE_HUMIDITY, 100),
  RH_min = pmin(Meteo_data$RELATIVE_HUMIDITY, 100),
  Rs_tot = Meteo_data$GLOBAL_RADIATION*3.6, # from kWh m-2 to MJ m-2
  uz_mean = Meteo_data$WIND_SPEED
)
```

```
head(ETo_data)
#> # A tibble: 6 x 2
#>   DATE      ETo
#>   <date>    <dbl>
#> 1 2020-01-01 0.226
#> 2 2020-01-02 0.418
#> 3 2020-01-03 0.651
#> 4 2020-01-04 0.407
#> 5 2020-01-05 0.561
#> 6 2020-01-06 0.518
```

Now we put everything in the AquacropOnR format:

```
Plu_example <- Meteo_data %>%
  dplyr::mutate(DAY = lubridate::yday(DATE), PLU = PRECIPITATION) %>%
  dplyr::select(DAY, PLU)
Tnx_example <- Meteo_data %>%
  dplyr::mutate(DAY = lubridate::yday(DATE), TMAX = TEMP_MAX, TMIN = TEMP_MIN) %>%
  dplyr::select(DAY, TMAX, TMIN)

ETo_example <- ETo_data %>%
  dplyr::mutate(DAY = lubridate::yday(DATE)) %>%
  dplyr::select(DAY, ETo)
```

Soil data

```
SOL_s <- tribble(~ID, ~Horizon, ~Thickness, ~SAT, ~FC, ~WP, ~Ksat, ~Penetrability, ~Gravel,
  "soil_1", 1, 0.1, 40.8, 29.2, 12.5, 120, 100, 0,
  "soil_1", 2, 0.2, 44.3, 32.2, 10.5, 96, 100, 0,
  "soil_1", 3, 0.6, 32.2, 21.4, 11.5, 52.5, 100, 0,
  "soil_1", 4, 1.1, 30.1, 20.2, 10.5, 55, 100, 0,
  "soil_2", 1, 0.1, 35.8, 29.2, 12.5, 1200, 100, 0,
  "soil_2", 2, 0.5, 31.8, 24.5, 10.5, 1200, 100, 0,
  "soil_2", 3, 2.2, 31.2, 22.2, 9.5, 950, 100, 0)

head(SOL_s)
#> # A tibble: 6 x 9
#>   ID      Horizon Thickness  SAT    FC    WP    Ksat Penetrability Gravel
#>   <chr>    <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>          <dbl> <dbl>
#> 1 soil_1      1        0.1  40.8  29.2  12.5  120          100      0
```

```
#> 2 soil_1      2      0.2 44.3 32.2 10.5 96      100      0
#> 3 soil_1      3      0.6 32.2 21.4 11.5 52.5    100      0
#> 4 soil_1      4      1.1 30.1 20.2 10.5 55      100      0
#> 5 soil_2      1      0.1 35.8 29.2 12.5 1200    100      0
#> 6 soil_2      2      0.5 31.8 24.5 10.5 1200    100      0
```

Initial conditions

If you do not provide initial conditions in the `design_scenario()` function, it will take the default setting “FC”, which results in a soil profile water content at field capacity and absence of salts at the start of the simulation.

The initial conditions need to have the same Horizon and Thickness variables as the soil input tibble `SOL_s`. here we provide an example where the initial water content at all layers lies in the middle between wilting point (WP) and field capacity (FC).

```
sw0_factor <- 0.5
SW0_s <- SOL_s %>% mutate(WC = WP + (FC-WP)*sw0_factor, ECe = 0.0) %>%
  select(ID, Horizon, Thickness, WC, ECe)
head(SW0_s)
#> # A tibble: 6 x 5
#>   ID      Horizon Thickness    WC    ECe
#>   <chr>      <dbl>     <dbl> <dbl> <dbl>
#> 1 soil_1      1      0.1  20.8     0
#> 2 soil_1      2      0.2  21.4     0
#> 3 soil_1      3      0.6  16.4     0
#> 4 soil_1      4      1.1  15.4     0
#> 5 soil_2      1      0.1  20.8     0
#> 6 soil_2      2      0.5  17.5     0
```

Management

Through the .MAN file, AquaCrop can take into account several field management settings, such as:

- the effect of mulching, using the percentage of soil covered by mulch (`mulch_perc`), and the effect on reducing soil evaporation (`mulch_eff` between 0 (no effect) and 100 (all evaporation inhibited)).
- the fertility stress (`fert_stress`, between 0 (no stress) and 100 (complete stress)).

- the presence of soil bunds (`soil_bunds`).
- the use of run-off affecting measures (`runoff_aff`, 0 or 1 (for complete prevention of run-off)) and their effect on the *curve number* (`CN_eff`).
- presence of weeds (`weed_clo`, `weed_mid`, `weed_cc`).

For an effect of erosion preventing measures, the management file can be defined as follows:

```
FMAN_s <- tribble(~ID, ~mulch_perc, ~mulch_eff, ~fert_stress, ~soil_bunds, ~runoff_aff, ~CN_eff,
  "default", 0, 50, 0, 0, 0, 0, 0, 0, 0, -0.01,
  "reduction_1", 0, 50, 0, 0, 0, -25, 0, 0, 0, -0.01)
```

where the *default* treatment has no measures and the *reduction_1* treatment has measures that reduce the curve number by 25.